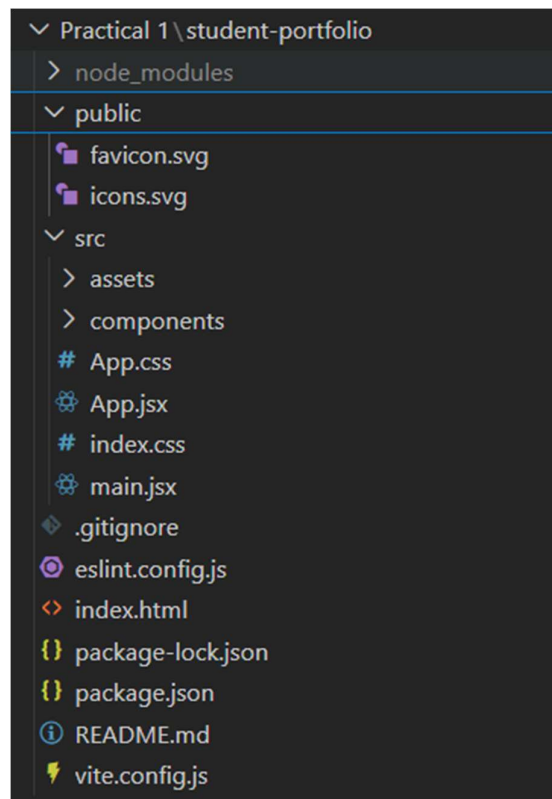


Practical 1 : Introduction to React and Component Architecture

Problem Definition :

- Create a React application using Vite for a student portfolio page.
- The application must include at least 4 reusable components: Header, About, Skills, and Footer.
- Each component must be independently structured and composed into a single-page layout.
- No logic or JSX should be duplicated across components.
- Props must be used to pass at least one piece of data into a minimum of 2 components.

Folder Structure :



about.jsx :

```
function About() {  
  return (  
    <section className="section">  
      <h2>About Me</h2>  
      <p>  
        my name is rudra and i am a student.  
      </p>  
    </section>  
  )  
}
```

```

    </p>
    <div className="details">
      <p><strong>College :</strong> DEPSTAR </p>
      <p><strong>Subject :</strong> ITUE301 - ADVANCED WEB DEVELOPMENT
FRAMEWORKS</p>
      <p><strong>Semester :</strong>5th Sem</p>
      <p><strong>Enrollment No :</strong> D25DIT081</p>
      <p><strong>Division :</strong> 5IT </p>
    </div>
  </section>
);
}

export default About;

```

header.jsx :

```

function Header(props) {
  return (
    <section className="hero">
      <h1 style={{ color: props.themeColor }}>{props.title}</h1>
      <h2>{props.name}</h2>
      <p>{props.role}</p>
      <button>View Skills</button>
    </section>
  );
}

export default Header;

```

navbar.jsx:

```

function Navbar() {
  return (
    <nav className="navbar">
      <h2 className="logo">Portfolio</h2>
      <ul>
        <li className="active">Home</li>
        <li>About</li>
        <li>Skills</li>
        <li>Contact</li>
      </ul>
    </nav>
  );
}

export default Navbar;

```

footer.jsx :

```

function Footer(props) {

```

```
    return (
      <section className="footer">
        <h2>Contact</h2>
        <p>Email : {props.email}</p>
        <p>Phone : {props.phone}</p>
        <p>City : {props.city}</p>
        <hr />
        <p>© 2026 Student Portfolio</p>
      </section>
    );
  }

export default Footer;
```

skills.jsx :

```
function Skills(props) {
  return (
    <section className="section">
      <h2>Technical Skills</h2>
      <div className="skills">
        {props.skillList.map((skill) => (
          <span className="skill" key={skill}>
            {skill}
          </span>
        ))}
      </div>
    </section>
  );
}

export default Skills;
```

App.jsx

```
import './App.css';

import Header from './components/Header';
import About from './components/About';
import Skills from './components/Skills';
import Footer from './components/Footer';
import Navbar from './components/Navbar';

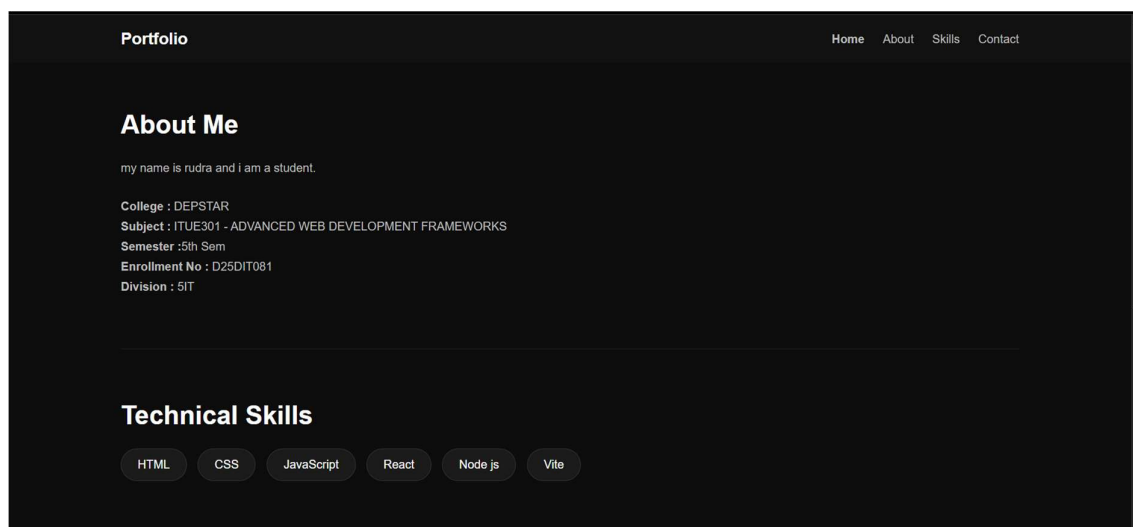
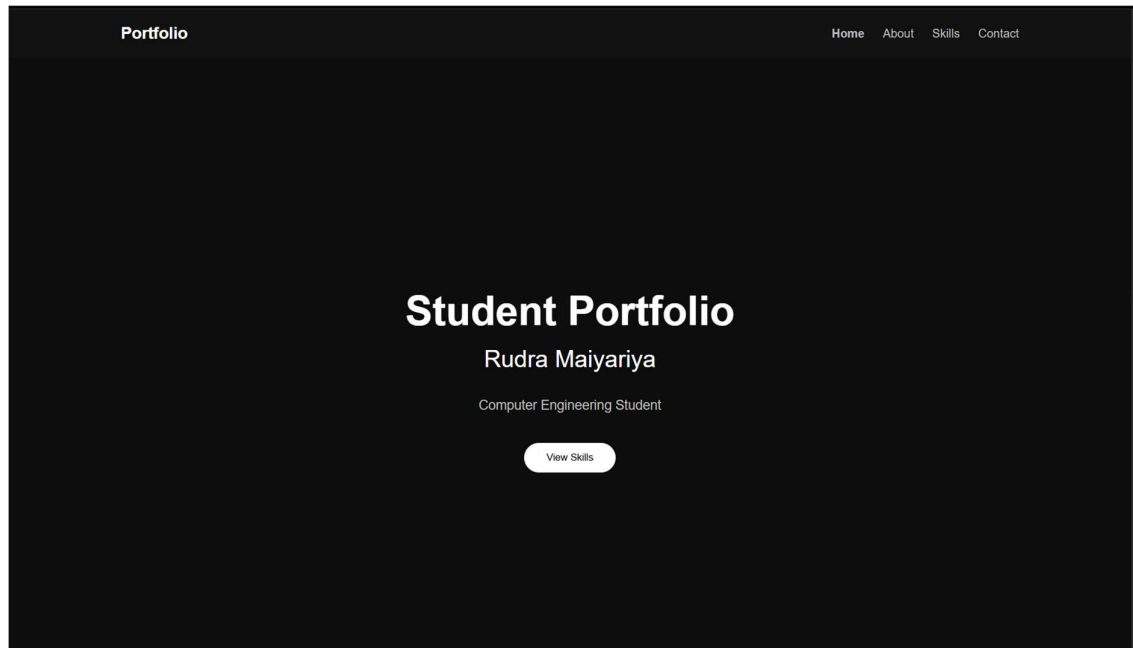
function App() {

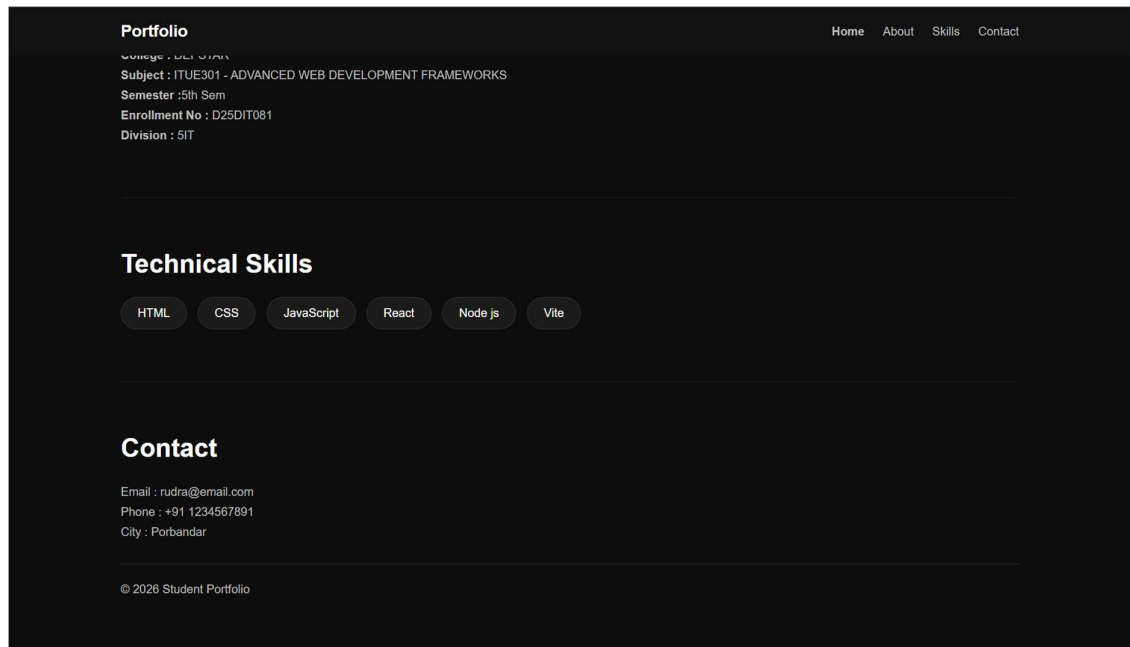
  const skills = [
    "HTML",
    "CSS",
    "JavaScript",
    "React",
    "Node js",
    "Vite"
  ];
```

```
return (  
  <div>  
    <Navbar />  
  
    <Header  
      title="Student Portfolio"  
      name="Rudra Maiyariya"  
      role="Computer Engineering Student"  
      themeColor="#ffffff"  
    />  
  
    <About />  
  
    <Skills skillList={skills} />  
  
    <Footer  
      email="rudra@email.com"  
      phone="+91 1234567891"  
      city="Porbandar"  
    />  
  
  </div>  
);  
}  
  
export default App;  
  
index.css  
  
body {  
  margin:0;  
}
```

Main.jsx:

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './index.css'  
import App from './App.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <App />  
  </StrictMode>,  
)
```

Output:



Supplementary Problems :

- Add a NavBar component that highlights the active section.

```
import React, { useState } from 'react';

function NavBar() {
  const [active, setActive] = useState('Home');
  const navItems = ['Home', 'About', 'Skills', 'Projects'];

  return (
    <nav style={{ display: 'flex', gap: '20px', padding: '20px', backgroundColor:
    '#f1f1f1' }}>
      {navItems.map(item => (
        <a
          key={item}
          href={`#${item.toLowerCase()}`}
          onClick={() => setActive(item)}
          style={{
            fontWeight: active === item ? 'bold' : 'normal',
            color: active === item ? '#0366d6' : '#333',
            textDecoration: 'none'
          }}
        >
          {item}
        </a>
      ))}
    </nav>
  );
}
```

```
}  
export default NavBar;
```

- **Pass an array of skills as a prop to the Skills component and render them dynamically.**

```
// Skills.jsx  
import React from 'react';  
  
function Skills({ skillsList }) {  
  return (  
    <section>  
      <h2>My Skills</h2>  
      <ul>  
        {skillsList.map((skill, index) => (  
          <li key={index}>{skill}</li>  
        ))}  
      </ul>  
    </section>  
  );  
}  
export default Skills;  
  
// Usage in App.jsx:  
// <Skills skillsList={['HTML', 'CSS', 'JavaScript', 'React']} />
```

- **Add a theme color prop to the Header and apply it as an inline style.**

```
// Header.jsx  
import React from 'react';  
  
function Header({ themeColor }) {  
  return (  
    <header style={{ backgroundColor: themeColor, padding: '20px', color: 'white' }}>  
      <h1>My Portfolio</h1>  
    </header>  
  );  
}  
export default Header;  
  
// Usage in App.jsx:  
// <Header themeColor="#282c34" />
```